Meta FAIR and the University of Edinburgh's new technique will verify the reasoning of LLMs

Meta FAIR (Fundamental AI Research) and the University of Edinburgh have unveiled Circuit-based Reasoning Verification (CRV), a groundbreaking white-box technique that can predict when a large language model's (LLM) reasoning is correct and intervene in real time to fix it. The method provides an interpretable view into an LLM's internal reasoning circuits, offering a major step towards transparent and debuggable AI.

Unlike 'black-box' approaches that evaluate only the final outputs, CRV inspects the model's reasoning circuits, specialised neuron subgraphs functioning like latent algorithms. By monitoring attribution graphs, which map how internal features influence tokens, CRV can trace specific reasoning flaws and correct them instantly. Trained transcoders replace dense transformer layers, creating interpretable computations and effectively installing a diagnostic port into the model.

When tested on the Llama 3.1 8B Instruct model across both synthetic and real-world datasets, CRV consistently outperformed traditional verification methods. The study also revealed domain-specific error signatures, showing that different reasoning tasks fail in distinct computational patterns. In one case, suppressing a prematurely firing 'multiplication' feature corrected an order-of-operations mistake mid-inference.

According to the research team, "shifting from opaque activations to interpretable computational structure enables a causal understanding of how and why LLMs fail to reason correctly."

AI coding agents from OpenAI, Anthropic, Google, Cognition, and xAI through a single 'mission control' interface. This marks a pivotal step in transforming GitHub into a unified control plane for AI-assisted software development, extending its tradition of openness into the AI era.

"GitHub has always done what it does best, fixing developer pain around how developers build, while still allowing you to choose how you want to build," said Kyle Daigle, chief operating officer, GitHub, underscoring the platform's enduring focus on developer autonomy and choice.

Third-party AI agents will roll out to GitHub Copilot subscribers in the coming months, with Copilot Pro+ users gaining early access to OpenAI Codex in VS Code Insiders. The move positions GitHub as a real-time AI 'command centre', enabling developers to control and monitor multiple agents simultaneously.

Perplexity unveils AI-driven patent research agent

Perplexity AI has launched Perplexity Patents, described by CEO Aravind Srinivas as "the world's first AI patent research agent." The tool is now available in free beta worldwide, offering users a simplified and conversational way to explore patent information using natural language queries rather than complex keyword syntax.

Designed to democratise intellectual property (IP) research, Perplexity Patents enables users to ask direct questions such as "Are there any patents on AI for language learning?" or "Key quantum computing patents since 2024?" It then delivers curated results with inline viewers and direct links to original documents.

The system supports follow-up questions, maintains context across searches, and suggests related topics, creating an interactive, AI-assisted research experience.

The platform stands out for its ability to detect conceptual similarities between documents, such as linking "fitness trackers" with "activity bands" or "health monitoring wearables." Beyond patent databases, Perplexity Patents integrates academic papers, open source software repositories, and other public innovation data, allowing users to track emerging technologies and research patterns early.

Perplexity Patents is currently free in beta for all users globally, with Pro and Max tiers offering expanded usage limits. The company's upcoming product, Perplexity Scholar, is aimed at bringing the same AI-powered accessibility to academic research.

IBM unveils Granite 4 Nano, a family of small generative AI models

IBM Corp. has released Granite 4 Nano, a family of compact, open source generative AI models, under the permissive Apache 2.0 licence, marking a major step towards decentralised and privacy-first AI. The models are designed to operate on-device, at the edge, or within browsers, enabling developers to deploy efficient and secure AI applications without cloud dependence.

The open source release allows unrestricted commercial use, modification, and redistribution, strengthening community-led innovation. By democratising access to sub-billion parameter models, IBM positions itself within the open LLM ecosystem alongside contributors like Alibaba's Qwen and Google's Gemma, advancing accessible AI for diverse environments.